

[← Go Back](#)

Hardware

MKR WiFi 1010

Tutorials

Debugging SAM-Based
Arduino® Boards with
Atmel-ICE

Accessing the Built-in RGB
LED on the MKR WiFi 1010

How to Connect Sensors to
the MKR WiFi 1010

Connecting MKR WiFi 1010
to a Wi-Fi Network

MKR WiFi 1010 Bluetooth®
Low Energy

Host a Web Server on the
MKR WiFi 1010

MKR WiFi 1010 Battery Application Note

Using the Segger J-Link
Debugger with the MKR
Boards

Sending Data over MQTT

Serial to OLED with MKR
WiFi 1010

Powering MKR WiFi 1010
with Batteries

RTC (Real Time Clock) with
MKR WiFi 1010 and OLED
Display

Scanning Networks with
MKR WiFi 1010

Securely Connecting an
Arduino MKR WiFi 1010 to
AWS IoT Core

Web Server Access Point
(AP) Mode with MKR WiFi
1010

Home / Hardware / MKR WiFi 1010 / **MKR WiFi 1010 Battery
Application Note**

MKR WiFi 1010 Battery Application Note

Learn about different topics related to batteries, and how to
choose the correct battery for you MKR application.

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Introduction

While you can run your MKR board from USB, you can
make best use of it when your MKR board is able to
operate independently of your computer and a USB cable.
This is particularly useful for many IoT applications, use of
relays with high current draw and remote sensing.
Batteries can help us in this regard and a special connector
is included in the MKR products for this end. In this
application note, we will take a closer look at the battery
capabilities of the MKR boards.

Goals

The goals of this project are to:

Understand what the terms **charging capacity** and
charging current mean.

Distinguish between LiPo and Li-Ion batteries

Learn how to identify connector polarity

Understand how the SAMD chip reads battery
voltage via a voltage divider

Hardware & Software Needed

Arduino MKR WiFi 1010

Multimeter

Arduino IDE

LiPo battery with JST-PH connector

ON THIS PAGE

Introduction

Goals

Hardware & Software
Needed

LiPo Vs Li-Ion Batteries

Connector

Protection Circuit

BQ24195L Power
Management IC

Electrical Properties

Multimeter Battery
Measurements

Read Battery Values Over
Serial

Conclusion

Further Ideas

time and are generally cheaper. Lithium Polymer batteries have a higher energy density, allowing you to run your board longer with a similar sized battery. You can see a comparison between these three in the table below.

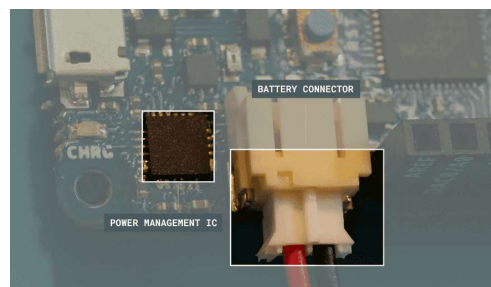
Battery Chemistry	Cost	Energy Density (mass)	Energy density (size)	Stability	Nominal voltage (single cell)
LiPo	Higher	higher	lower	Higher	3.7V
Li-ion	Lower	Lower	Higher	Lower	3.7V

i Warning: Please make sure that the battery terminals do NOT touch each other. The short circuit can cause permanent damage to the battery and even explosion.

Apart from being a single cell (3.7V), a key factor to selecting the correct battery is that the connector is compatible.

Connector

You can connect a battery to the MKR WIFI 1010 via a 2-pin JST-PH female connector. The PH variant of JST connectors are identified by a pin-to-pin distance of 2 mm. Here are several examples of LiPo batteries with a 2-pin JST-PH connector. Each individual connector is made of one plastic housing and two metal crimp terminals. A crimping device may be required. Note that when looking from above (with the notch facing you), the red (positive) wire should be on your left.



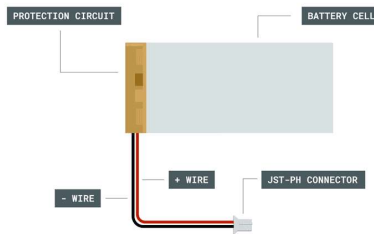
JST-PH connector connected to the MKR WiFi 1010. Note that the positive terminal (red wire) is to the left, towards the PMIC

You can connect a battery to the MKR WIFI 1010 via a 2-pin JST-PH female connector. The PH variant of JST connectors are identified by a pin-to-pin distance of 2mm. Here are several examples of LiPo batteries with a 2-pin JST-PH connector. Each individual connector is made of one plastic housing and two metal crimp terminals. A crimping device may be required. Note that when looking from

above (with the notch facing you), the red (positive) wire should be on your left.

Protection Circuit

A protection circuit cuts off the battery if overcurrent or under/over voltage is detected. This adds an additional layer of safety.



LiPo Battery Breakdown

BQ24195L Power Management IC

The BQ24195L IC is responsible for charging the single cell lion battery, as well as boosting the battery voltage to 3.3V for use on the board internally. An Arduino library ([Arduino_BQ24165](#)) has been developed, to help control the features of this battery over I2C.

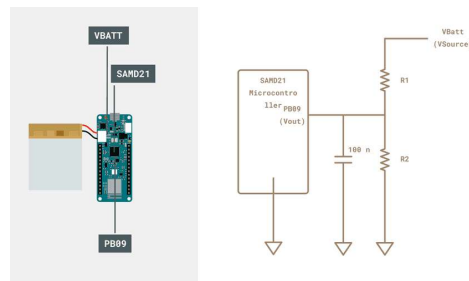
Electrical Properties

Voltage The nominal voltage of both LiPo and Li-Ion batteries is around 3.7V and is commonly referred to as the voltage of single cell batteries. However, this is an ideal scenario since as a battery is used the voltage changes. The voltage can be considered similar to a waterfall. The higher the waterfall, the higher the voltage.

In the MKR boards, the battery terminal is connected to the SAMD21 via a reserved pin (PB09) known as `ADC_BATTERY` within the Arduino Core. This pin is used internally on the board, and is not accessible via the MKR pins. Since the voltage of a Li-ion battery exceeds 3.3V (the AREF value), a voltage divider must be used to extend the range while also ensure that only safe voltages are applied to the microcontroller. We can calculate the output voltage using the following formula

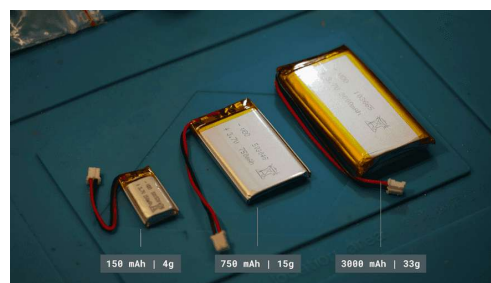
$$V_{out} = \frac{V_{source} \times R_2}{R_1 + R_2}$$

In the MKR WIFI 1010, R_1 and R_2 are 330k ohm and 1M ohm respectively. Therefore, for a resolution of 12 bits, the board is subject to 3.3V that corresponds to about 4.39V on the battery side. Therefore, we can cover the operating value of a whole battery. The high values reduce the leakage current that may pass through, increasing the life of the battery. The capacitor acts to clean the signal.



Voltage Divider Circuit on the MKR WIFI 1010

Capacity Continuing with the waterfall analogy, the volume of water passing through a waterfall represents the current. Therefore, the amount of water stored behind the waterfall is considered the capacity. It is common to discuss the capacity in terms of milliamp hours (mAh), which is the current that can be potentially extracted in an hour to discharge it. Note that changes in temperature and elevated current demands can change the effective capacity of your battery.



LiPo batteries with various capacity sizes

While the MKR boards do not provide a mechanism for identifying capacity, we can get a general idea of the current status by mapping the voltage to the capacity. A

gauge IC (Integrated Circuit), available in the Portenta X8 and H7.

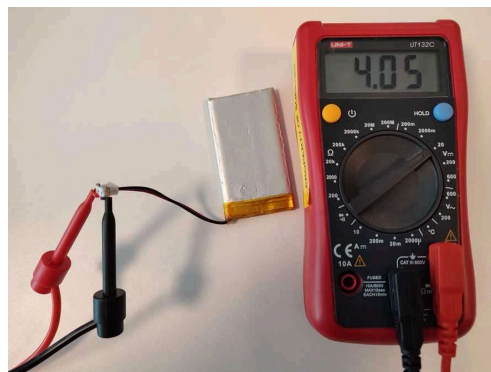
Discharge rating When the battery is fully charged or when it is near to discharge, the discharge rate changes considerably. Yet, there is a region where the discharge rate is constant (change in voltage over change in discharge capacity does not fluctuate). Within this region, the maximum current draw is defined as:

$$\begin{aligned} \text{maximum current draw} \\ = \text{battery capacity} \times \text{discharge rating} \end{aligned}$$

The discharge rating (C) is often provided in the datasheet of the battery. If the C rating of a battery is 1, then it can discharge the maximum current for one hour before running out. As a rule of thumb, higher discharge rates lead to the reduction of the effective capacity and lifetime of the battery.

Multimeter Battery Measurements

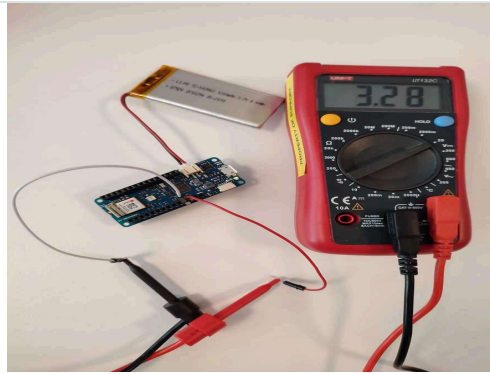
1. Test the battery voltage with a multimeter. In this case, we have used a 1200mAh battery. It should be between 3.3V and 4.2V, regardless of the capacity. If the battery voltage is outside this range, your battery may be damaged.



Multimeter connected to a LiPo battery

i Warning: Please make sure that the battery terminals do NOT touch each other. The short circuit can cause permanent damage to the battery and even explosion.

2. Connect your battery to the MKR WIFI 1010 then check



Measurement of VCC when LiPo battery is connected

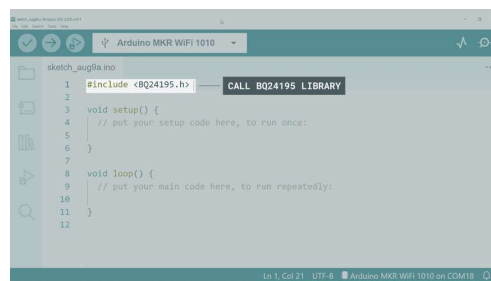
You should notice that the voltage of VCC is about 3.3V, regardless of the battery level. The voltage that comes out of the PMIC enters the VCC.

Read Battery Values Over Serial

We will go through the lines needed to create a Sketch to read the battery values over Serial and give a short description of what each part does. At the end, the full code will be provided so you can copy and paste it into your IDE and upload it to your Arduino board.

1. Open a new sketch in the Arduino IDE. We will create a sketch to read the ADC voltage that is sensed by the SAMD controller. As a first step, we will call the library to be included in the sketch.

```
1 #include <BQ24195.h>
```



Calling the BQ24195 Library

2. Then, we will create variables to store the variables for the raw ADC value (from pin PB09), the equivalent voltage experienced by PB09 and finally the calculated battery voltage.

```

1 float rawADC;
2   float voltADC;
3   float voltBat;

```

3. According to the schematics for the MKR WIFI 1010, we will now define the values for the R1 and R2 resistor. These will be used to calculate `voltBat`.

```

1 int R1 = 330000;
2   int R2 = 1000000;

```

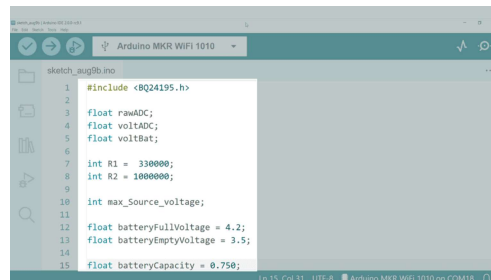
4. We will now create a variable to store the maximum source voltage `max_Source_voltage` as well as the upper (`batteryFullVoltage`) and lower (`batteryEmptyVoltage`) values for the battery. We will also define the battery capacity as `batteryCapacity` so as to determine the charging current. Since we are using a 750 mAh battery in this example, we will set the value to `0.750`.

```

1 int max_Source_voltage;
2
3   float batteryFullVoltage = 4.2;
4   float batteryEmptyVoltage = 3.3;
5
6   float batteryCapacity = 0.750;

```

5. Your IDE should look similar to the screenshot below.



Initial library & variable configuration code

6. Now we can configure the `setup()` function. We need to initiate the Serial connection (`Serial.begin(9600)`), ensure the analog reference is set to the default value of 3.3V (`analogReference(AR_Default)`) and set the ADC resolution to 12 bits (`analogReadResolution(12)`). A 12bit ADC will provide an output from 0 (0V) to 4093 (3.3V), depending on the voltage the ADC experiences.

```
1 void setup() {  
2   Serial.begin(9600);  
3  
4   analogReference(AR_DEFAULT);  
5   analogReadResolution(12);
```

7. Now we can establish the SAMD21-BQ24195 connection over I2C and enable the boost mode. By default, the PMIC provides an output close to 3.3V to reduce the power loss over the 3.3V linear regulator. By enabling boost mode, we can access 5V on the VCC pin.

```
1 PMIC.begin();  
2 PMIC.enableBoostMode();
```

8. Here, we will define the minimum output voltage of the PMIC as well as the maximum level to which the battery is charged.

```
1 PMIC.setMinimumSystemVoltage(batteryEmptyVoltage);  
2 PMIC.setChargeVoltage(batteryFullVoltage);
```

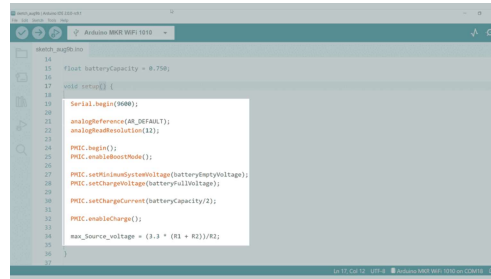
9. As a rule of thumb, we can now set the maximum charging current to half of the battery capacity. With the configurations made for the battery, we can now enable the charging circuit.

```
1 PMIC.setChargeCurrent(batteryCapacity/2);  
2  
3 PMIC.enableCharge();
```

10. We will add a final line to the `setup()` function that will calculate the voltage that when applied to the input of the voltage divider will yield 3.3V at PB09. This is the upper limit for the voltage sensing capabilities.

```
1 max_Source_voltage = (3.3 * (R1 + R2))/R2;  
2 }
```

11. Your `setup()` function should now look as follows.



Serial & BQ24196 Setup

12. With the variables and setup function clearly defined, we will now specify the loop function. These commands will continuously run and provide information about the battery status to the Serial Monitor in the PC. First, we read the value of PB09 (specified internally as `ADC_BATTERY`). The value represented by `rawADC` is a number between 0 to 4094, given that it has 12 bit resolution.

```

1 void loop()
2 {
3
4   rawADC = analogRead(ADC_BATTERY);

```

13. In order to convert `rawADC` into a voltage reading (`voltADC`) we will divide `rawADC` by 4095 and then multiply it by the analog reference voltage (3.3V).

```

1 voltADC = rawADC * (3.3/4095.0);

```

14. The `voltADC` variable gives us the voltage sensed directly on the PB09 pin. This voltage is passed through the voltage divider, so it is a fraction of the actual battery voltage. We can then calculate the equivalent battery voltage as follows.

```

1 voltBat = voltADC * (max_source_voltage/3.3);

```

15. We can approximate the battery voltage to be proportional to the capacity level. Since the `map()` function does not work with float variables, we will manually map the values.

```
1 int new_batt = (voltBat - batteryEmptyVolta
```

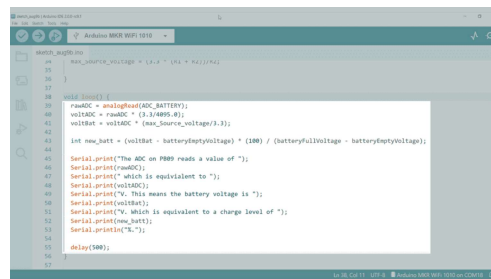
16. We can now send the obtained values over Serial.

```
1 Serial.print("The ADC on PB09 reads a value
2   Serial.print(rawADC);
3   Serial.print(" which is equivalent to ")
4   Serial.print(voltADC);
5   Serial.print("V. This means the battery v
6   Serial.print(voltBat);
7   Serial.print("V. Which is equivalent to a
8   Serial.print(new_batt);
9   Serial.println("%.");
```

17. Finally, we will add a half a second delay at the end of the `loop()` function to allow variables to come slowly through the Serial Monitor.

```
1 delay(500);
2 }
```

18. Your section for the `loop()` function should look as follows.



Battery level calculation and Serial printing

19. The complete code (with the addition of comments) is as follows. You can copy and paste this directly into your IDE

```

1  /*
2   Read battery voltage on MKR WIFI 1010 ;
3
4
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6
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25
26   Author: Ali Jahangiri & Karl Soderby
27
28   Last Edit: 9th August 2022

```

12. Connect the LiPo battery to your MKR WIFI 1010, without the USB cable connected.



The battery must be connected to the MKR WIFI 1010 before connecting via USB, otherwise it may not work.

13. Connect the MKR WIFI 1010 to the USB port of your PC. Make sure that the Arduino MKR WIFI 1010 is selected and that the port is correct. Upload the code to the board and open the terminal monitor. You will see information about the battery level displayed in the Serial monitor

Conclusion

In this Application Note, you have gained an understanding of how the batteries connect to the MKR WiFi 1010, both physically via the JST-PH connector as well as the I2C-controlled BQ24195 IC. You can now select the correct battery capacity and configure it for your application.

Further Ideas

Try disabling the charging circuit with

Try experimenting with different battery capacities, and see how long it can power your application.

Send a just the `voltBat` over the serial and use the Serial Plotter to view the values as a graph.

Suggest changes

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